

Dominique Schröder

Professor of Privacy-Enhancing Technologies at TU Wien
schroeder@me.com

Academic appointments

Technische Universität Wien, Vienna, Austria Full Professor (Univ. Prof. with tenure)	2024-present
Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany Full Professor (W3, with tenure)	2016-2024
Saarland University, Germany Professor (W2, with tenure)	08/2015-06/2016
Saarland University, Germany Assistant Professor (W1, tenure-track)	08/2012-07/2015
University of Maryland, United States of America Postdoctoral Researcher under Jonathan Katz	02/2011-06/2012
Technical University of Darmstadt, Germany Doctoral Student under Marc Fischlin	10/2006-01/2011

Education

Technical University Darmstadt Dr. rer. nat. (with distinction) · Cryptography	10/2006-11/2010
Technical University Braunschweig Diplom Informatiker. (Diploma/M.Sc.) · Computer Science · Identification Schemes	10/2002-09/2006

Research interests

How can we compute on private data and share control of digital systems without trusting a single party?

I design cryptographic protocols that distribute trust and protect sensitive information. My research asks how their security guarantees hold up when participants are compromised or protocols become part of larger systems.

Honors and awards

Feodor Lynen Award Humboldt Foundation	2021
Efficient Unlinkable Sanitizable Signatures from Signatures with Re-Randomizable Keys Distinguished Paper · Invited from PKC 2016 to IET IFS Special Issue	2017
Delegtable Functional Signatures Distinguished Paper · Invited from PKC 2016 to IET IFS Special Issue	2017

(Efficient) Universally Composable Oblivious Transfer Using a Minimal Number of Stateless Tokens Distinguished Paper · Invited from TCC 2014 to Journal of Cryptology	2014
Intel Early Career Faculty Award Intel · Awarded to 6 people in Europe (\$35,000)	2013
Postdoctoral Fellowship Deutscher Akademischer Austausch Dienst (DAAD) · University of Maryland	2013
Dissertation TU Darmstadt · summa cum laude (with highest honors)	2012
Research Fellow Deutscher Akademischer Austausch Dienst (DAAD) · Research visit at the University of Maryland	2010
Best Teaching Assistant Award TU Darmstadt · Lecture Introduction to Modern Cryptography	2007
Programming Contest N@tkids Winner of the programming contest “N@tkids” · akte 2000 and ricardo.de in the third category (5000 DM). Invited to the TV show akte 2000.	2000

Research visits

Visiting Professor · University of Maryland, USA Host: Prof. Katz	07/2023-09/2023
Visiting Professor · University of Berkeley, USA Host: Prof. Garg	08/2016
Visiting Professor · Chinese University of Hong Kong, Hong Kong Host: Prof. Chow	03/2016
Visiting Professor · University of Maryland, USA Host: Prof. Katz	07/2015-08/2015
Visiting Professor · University of Maryland, USA Host: Prof. Katz	08/2013-10/2013
Research Internship · University of Maryland, USA Host: Prof. Katz	07/2010-10/2010

Research funding and projects

Unrestricted Gift Google · sole PI · 120.000 EUR	10/2026-12/2028
Foundations and Applications of Resource-Restricted Cryptography WWTF · PI · 299.708 EUR	01/2026-12/2029
Confidential 6G EU Horizon · PI · 282.500 EUR	01/2023-12/2026
Post-quantum Secure Key-Exchange AUDI · Research Contract · 100.000 EUR	04/2022-03/2023
Password-based Cryptography DFG · sole PI · 327.000 EUR	04/2021-03/2023

NCT: Privacy Enhancing Technologie State of Bavaria · PI · 351.000 EUR	10/2021-09/2025
Privacy-Preserving Blockchains Protocol Labs · PI · 23.000 EUR	10/2023-09/2024
Postquantum Security for Automobiles AUDI · Research Contract · 60.000 EUR	01/2021-12/2021
Cybercrime and Forensic Computing DFG Integrated Research Training Group · PI · 645.000 EUR	08/2019-07/2028
PROMISE - Personal Medial Safe BMBF · Leading PI · 710.000 EUR	01/2016-12/2019
Password-based Cryptography DAAD · PI · 14.000 EUR	01/2018-12/2019
Methods and Tools for Understanding Privacy DFG - Collaborative Research Center (SFB) · PI · 325.000 EUR	01/2016-12/2019
CISPA Center for IT-Security, Privacy, and Accountability BMBF · PI · 690.000 EUR	01/2013-04/2018
Cryptographic Protocols for Cloud Storage DAAD · sole PI · 12.000 EUR	01/2015-12/2016
Cryptography for Secure Digital Interaction EU COST · PI · 135.000 EUR	11/2013-04/2018
Usable Security for Mobile Devices BMBF · PI · 100.000 EUR	01/2014-12/2014
TWC Small: Exploring Cryptographic Models NSF · PI, but I left to Germany · 314.000 EUR	09/2012-08/2015

Publications

BitPriv: A Privacy-Preserving Protocol for DeFi Applications on Bitcoin Ioannis Alexopoulos, Zeta Avarikioti, Paul Gerhart, Matteo Maffei, and Dominique Schröder	NDSS 2027
Fully Adaptive FROST with Identifiable Aborts from AOMDL Ruben Baecker, Paul Gerhart, Davide Li Calsi, Luigi Russo, Dominique Schröder, and Arkady Yerukhimovich	ASIACRYPT 2026
Fair Distributed Exchange via Threshold Adaptor Signatures Ruben Baecker, Paul Gerhart, Jonathan Katz, and Dominique Schröder	ASIACRYPT 2026
Universally Composable Adaptor Signatures Paul Gerhart, Daniel Rausch, and Dominique Schröder	ACM CCS 2026
Fully-Adaptive Two-Round Threshold Schnorr Signatures from DDH Paul Gerhart, Davide Li Calsi, Luigi Russo, and Dominique Schröder	EUROCRYPT 2026
Password-Hardened Encryption Revisited Ruben Baecker, Paul Gerhart, and Dominique Schröder	ASIACRYPT 2025

Universally Composable and Round-Optimal Password-Hardened Encryption Behzad Abdolmaleki, Ruben Baecker, Paul Gerhart, Mike Graf, Mojtaba Khalili, Daniel Rausch, Fritz Schmidt, and Dominique Schröder	ASIACRYPT 2025
A Fully-Adaptive Threshold Partially-Oblivious PRF Ruben Baecker, Paul Gerhart, Daniel Rausch, Dominique Schröder	CRYPTO 2025
Increasing the Resilience of Secure Multiparty Computation using Security Modules Lamaya Omar, Felix Freiling, Dominique Schröder	Transactions on Dependable and Secure Computing 2025
Automated Analysis and Synthesis of Message Authentication Codes Stefan Milius, Dominik Paulus, Julian Thomas, Dominique Schröder, and Lutz Schröder	CSF 2025
SoK: Descriptive Statistics Under Local Differential Privacy René Raab, Pascal Berrang, Paul Gerhart, and Dominique Schröder	PETS 2025
Measuring Conditional Anonymity - A Global Study Pascal Berrand, Paul Gerhart, and Dominique Schröder	PETS 2024
Foundations of Adaptor Signatures Paul Gerhart, Dominique Schröder, Pratik Soni, and Sri A. K. Thyagarajan	EUROCRYPT 2024
Practical Schnorr Threshold Signatures without the Algebraic Group Model Hien Chu, Paul Gerhart, Tim Ruffing, Dominique Schröder	CRYPTO 2023
ROAST: Robust Asynchronous Schnorr Threshold Signatures Tim Ruffing, Viktoria Ronge, Elliott Jin, Jonas Schneider-Bensch, and Dominique Schröder	ACM CCS 2022
Verifiable Timed Linkable Ring Signatures For Scalable Payments for Monero Sri A. K. Thyagarajan, Giulio Malavolta, Fritz Schmidt, Dominique Schröder	ESORICS 2022
Inner Product Functional Commitments with Constant-Size Public Parameters and Openings Hien Chu, Dario Fiore, Dimitris Kolonelos, and Dominique Schröder	SCN 2022
Everlasting UC Commitments from Fully Malicious PUFs Bernardo Magri, Giulio Malavolta, Dominique Schröder, and Dominique Unruh	Journal of Cryptology 2022
A Security Framework for Distributed Ledgers Mike Graf, Daniel Rausch, Viktoria Ronge, Christoph Egger, Ralf Küsters, and Dominique Schröder	ACM CCS 2021
OpenSquare: Decentralized Repeated Modular Squaring Service Sri A. K. Thyagarajan, Adithya Bhat, Tiantian Gong, Aniket Kate, and Dominique Schröder.	ACM CCS 2021

- CoinJoin in the Wild - An Empirical Analysis in Dash** ESORICS
2021
Dominic Deuber and Dominique Schröder
- Foundations of Ring Sampling** PETS
2021
Viktoria Ronge, Christoph Egger, Russell W. F. Lai, Dominique Schröder,
Hoover H. F. Yin
- Controlling My Genome With My Smartphone a First Clinical Experiences
Of The PROMISE-System** Clinical Research in
Cardiology
2021
Ali Amr, Marc Hinderer, Lena Griebel, Dominic Deuber, Christoph Egger,
Farbod Sedaghat-Hamedani, Elham Kayvanpour, Daniel Huhn, Jan Haas, Karen
Frese, Marc Schweig, Ninja Marnau, Annika Krämer, Claudia Durand, Florian
Battke, Hans-Ulrich Prokosch, Michael Backes, Andreas Keller, Dominique
Schröder, Hugo A. Katus, Norbert Frey, Benjamin Meder
- Threshold Password-Hardened Encryption Services** ACM CCS
2020
Julian Brost, Christoph Egger, Russell W. F. Lai, Fritz Schmid, Dominique
Schröder, and Markus Zoppelt
pp. 409-424
- Verifiable Timed Signatures Made Practical** ACM CCS
2020
Sri A. K. Thyagarajan, Adithya Bhat, Giulio Malavolta, Nico Döttling, Aniket Kate,
and Dominique Schröder
pp. 1733-1750
- The Patient as Genomic Data Manager - Evaluation of the PROMISE App.** MIE
2020
Lena Griebel, Marc Hinderer, Ali Amr, Benjamin Meder, Marc Schweig, Dominic
Deuber, Christoph Egger, Claudia Kawohl, Annika Krämer, Isabell Flade,
Dominique Schröder, Hans-Ulrich Prokosch
pp. 1061-1065
- Feasibility and Infeasibility of Secure Computation with Malicious PUFs** Journal of Cryptology
2020
Dana Dachman-Soled, Nils Fleischhacker, Jonathan Katz, Anna Lysyanskaya,
and Dominique Schröder
pp. 595-617
- Omniring: Scaling Up Private Payments Without Trusted Setup - Formal
Foundations and Constructions of Ring Confidential Transactions with
Log-size Proofs** ACM CCS
2019
Russell W. F. Lai, Viktoria Ronge, Tim Ruffing, Dominique Schröder, Sri A. K.
Thyagarajan, and J.Wang
pp. 31-48
- Efficient Invisible and Unlinkable Sanitizable Signatures** PKC
2019
Xavier Bultel, Pascal Lafourcade, Rusell W.F. Lai, Giulio Malavolta, Dominique
Schröder, and Sri A. K. Thyagarajan
pp. 159-189
- My Genome Belongs to Me: Controlling Third Party Computation on
Genomic Data** PETS
2019
Dominic Deuber, Christoph Egger, Katharina Fech, Giulio Malavolta, Sri A. K.
Thyagarajan, Florian Battke, Claudia Durand, and Dominique Schröder
pp. 108-132

- Tight Security Proofs for Schnorr Signatures**
Nils Fleischhacker, Tibor Jager, and Dominique Schröder
pp. 566-599
Journal of Cryptology 2019
- (Efficient) Universally Composable Oblivious Transfer Using a Minimal Number of Stateless Tokens**
Seung Geol Choi, Jonathan Katz, Dominique Schröder, Arkady Yerukhimovich, Hong-Sheng Zhou
pp. 459-497
Journal of Cryptology 2019
- Group ORAM for privacy and access control in outsourced personal records**
Matteo Maffei, Giulio Malavolta, Manuel Reinert, Dominique Schröder
pp. 1-47
Journal of Computer Security 2019
- Homomorphic Secret Sharing for Low Degree Polynomials**
Russell W. F. Lai, Giulio Malavolta, and Dominique Schröder
pp. 279-309
ASIACRYPT 2018
- Simple Password-Hardened Encryption Services**
Russell W. F. Lai, Christoph Egger, Manuel Reinert, Sherman S. M. Chow, Matteo Maffei, and Dominique Schröder
pp. 1405-1421
USENIX 2018
- Functional Credentials**
Dominic Deuber, Matteo Maffei, Giulio Malavolta, Max Rabkin, Dominique Schröder, Mark Simkin.
pp. 64-84
PETS 2018
- Burning Zerocoins for Fun and for Profit: A Cryptographic Denial-of-Spending Attack on the Zerocoin Protocol**
Tim Ruffing, Sri A. K. Thyagarajan, Viktoria Ronge, Dominique Schröder
pp. 116-119
Crypto Valley Conference on Blockchain Technology 2018
- Subset Predicate Encryption and Its Applications**
Jonathan Katz, Matteo Maffei, Giulio Malavolta, and Dominique Schröder
CANS 2017
- On the Security of Frequency-Hiding Order-Preserving Encryption**
Matteo Maffei, Manuel Reinert, and Dominique Schröder
pp. 115-134
CANS 2017
- Efficient Ring Signatures in the Standard Model**
Giulio Malavolta and Dominique Schröder
pp. 128-157
ASIACRYPT 2017
- Phoenix: Rebirth of a Cryptographic Password-Hardening Service**
Russell W.F.Lai, Christoph Egger, Dominique Schröder, and Sherman S.M.Chow
pp. 899-916
USENIX 2017
- Maliciously Secure Multi-Client ORAM**
Matteo Maffei, Giulio Malavolta, Manuel Reinert, and Dominique Schröder
pp. 645-664
ACNS 2017

- Security of Blind Signatures Revisited
Dominique Schröder and Dominique Unruh
pp. 470–494
Journal of Cryptology
2017
- Efficient Cryptographic Password Hardening Services From Partially Oblivious Commitments
Jonas Schneider, Nils Fleischhacker, Dominique Schröder, and Michael Backes
pp. 1192–1203
ACM CCS
2016
- Efficient Sanitizable Signatures without Random Oracles
Russell W. F. Lai, Tao Zhang, ShermanChow and Dominique Schröder
pp. 363–380
ESORICS
2016
- Two Message Oblivious Evaluation of Cryptographic Functionalities
Nico Doettling, Nils Fleischhacker, Johannes Krupp, and Dominique Schröder
pp. 619–648
CRYPTO
2016
- Delegtable Functional Signatures
Michael Backes, Sebastian Meiser, and Dominique Schröder
pp. 357–386
PKC
2016
- Efficient Unlinkable Sanitizable Signatures from Signatures with Re-Randomizable Keys
Nils Fleischhacker, Johannes Krupp, Giulio Malavolta, Jonas Schneider, Dominique Schröder, and Mark Simkin
pp. 301–330
PKC
2016
- Nearly Optimal Verifiable Data Streaming
Johannes Krupp, Dominique Schröder, Mark Simkin, Dario Fiore, Giuseppe Ateniese, and Stefan Nuernberger
pp. 417–445
PKC
2016
- Efficient Unlinkable Sanitizable Signatures from Signatures with Re-Randomizable Keys
Nils Fleischhacker, Giulio Malavolta, Dominique Schröder, and Mark Simkin
pp. 166–183
IET IFS Special Issue
2016
- Delegtable Functional Signatures
Sebastian Meiser and Dominique Schröder.
pp. 194–206
IET IFS Special Issue
2016
- Efficient Pseudorandom Functions via On-the-Fly Adaptation
Nico Döttling and Dominique Schröder
pp. 329–350
CRYPTO
2015
- Privacy and Access Control for Outsourced Personal Records
Matteo Maffei, Giulio Malavolta, Manuel Reinert, and Dominique Schröder
pp. 341–358
IEEE Symposium on Security and Privacy
2015
- Liar, Liar, Coins on Fire! Penalizing Equivocation By Loss of Bitcoins
Tim Ruffing, Aniket Kate, and Dominique Schröder
pp. 219–230
ACM CCS
2015

Verifiably Encrypted Signatures: Security Revisited and a New Construction Christian Hanser, Max Rabkin, and Dominique Schröder	ESORICS 2015
VeriStream – A Framework for Verifiable Data Streaming Dominique Schröder and Mark Simkin pp. 146-164	FC 2015
Foundations of Reconfigurable PUFs Jonas Schneider and Dominique Schröder pp. 579-594	ACNS 2015
Feasibility and Infeasibility of Secure Computation with Malicious PUFs Dana Dachman-Soled, Nils Fleischhacker, Jonathan Katz, Anna Lysyanskaya, and Dominique Schröder pp. 405-420	CRYPTO 2014
Tight Security Proofs for Schnorr Signatures Nils Fleischhacker, Tibor Jager, and Dominique Schröder pp. 512-531	ASIACRYPT 2014
(Efficient) Universally Composable Oblivious Transfer Using a Minimal Number of Stateless Tokens Seung Geol Choi, Jonathan Katz, Dominique Schröder, Arkady Yerukhimovich, Hong-Sheng Zhou pp. 638-662	TCC 2014
Brief Announcement: Towards Security and Privacy for Outsourced Data in the Multi-Party Setting Matteo Maffei, Giulio Malavolta, Manuel Reinert, and Dominique Schröder pp. 144-146	PODC 2014
Demo: Security and Privacy with Google Glass Johannes Krupp, Dominique Schröder, and Mark Simkin pp. 1445-1447	CCS 2014
Ubic: Bridging the gap Between Digital Cryptography and the Physical World Mark Simkin, Dominique Schröder, Andreas Bulling, and Mario Fritz pp. 56-75	ESORICS 2014
WebTrust – A Comprehensive Authenticity and Integrity Framework for HTTP Michael Backes, Reiner W. Gerling, Sebastian Gerling, Stefan Nürnberger, Dominique Schröder, and Mark Simkin pp. 401-418	ACNS 2014
Verifiable Data Streaming Dominique Schröder and Heike Schröder pp. 953-964	CCS 2012

Uniqueness is a Different Story: Impossibility of Verifiable Random Functions from Trapdoor Permutations Dario Fiore and Dominique Schröder pp. 636-653	TCC 2012
Security of Blind Signatures Revisited Dominique Schröder and Dominique Unruh pp. 662-679	PKC 2012
History-Free Sequential Aggregate Signatures Marc Fischlin, Anja Lehmann, and Dominique Schröder pp. 113-130	SCN 2012
Security of Blind Signatures Under Aborts and Applications to Adaptive Oblivious Transfer Marc Fischlin and Dominique Schröder. pp. 169-204	Journal Mathematical Cryptology 2012
Round Optimal Blind Signatures Dominique Schröder and Dominique Unruh pp. 630-648	CRYPTO 2011
Impossibility of Blind Signatures from One-Way Permutations Jonathan Katz, Dominique Schröder, Arkady Yerukhimovich pp. 615-629	TCC 2011
How to Aggregate the CL Signature Dominique Schröder pp. 298-314	ESORICS 2011
Expedient Non-Malleability Notions for Hash Functions Paul Baecher, Marc Fischlin, and Dominique Schröder pp. 268-283	CT-RSA 2011
On the Impossibility of Three-Move Blind Signature Schemes Marc Fischlin and Dominique Schröder pp. 197-215	EUROCRYPT 2010
Unlinkability of Sanitizable Signatures Christina Brzuska, Marc Fischlin, Anja Lehmann, and Dominique Schröder pp. 444-461	PKC 2010
Confidential Signatures and Deterministic Signcryption Alex W. Dent, Marc Fischlin, Mark Manulis, Dominique Schröder, and Martijn Stam pp. 462-479	PKC 2010

Redactable Signatures for Tree-Structured Data: Definitions and Construction Christina Brzuska, Heike Busch, Özgür Dagdelen, Marc Fischlin, Martin Franz, Stefan Katzenbeisser, Mark Manulis, Cristina Onete, Andreas Peters, Bertam Poettering, and Dominique Schröder pp. 87-104	ACNS 2010
Generic Constructions for Verifiably Encrypted Signatures without Random Oracles or NIZKs Markus Rückert, Michael Schneider, and Dominique Schröder pp. 69-86	ACNS 2010
History-Free Aggregate Message Authentication Codes Oliver Eikemeier, Marc Fischlin, Jens-Fabian Götzmann, Anja Lehmann, Dominique Schröder, Peter Schröder, and Daniel Wagner pp. 309-328	SCN 2010
Fair Partially Blind Signatures Markus Rückert and Dominique Schröder pp. 34-51	Africacrypt 2010
Public-Key Encryption with Non-interactive Opening: New Constructions and Stronger Definitions David Galindo, Benoit Libert, Marc Fischlin, Georg Fuchsbauer, Anja Lehmann, Mark Manulis, and Dominique Schröder pp. 333-350	Africacrypt 2010
CAPTCHAs: The Good, the Bad, and the Ugly. Paul Baecher, Lior Gordon, Marc Fischlin, Michael Luetzow, Dominique Schröder pp. 353-365	Sicherheit 2010
Security of Blind Signatures Under Aborts Marc Fischlin and Dominique Schröder pp. 117-128	PKC 2009
Security of Sanitizable Signatures Revisited Christina Brzuska, Marc Fischlin, Tobias Freudenreich, Anja Lehmann, Marcus Page, Jakob Schelbert, Dominique Schröder, and Florian Volk pp. 297-316	PKC 2009
Security of Verifiably Encrypted Signatures and a Construction Without Random Oracles Markus Rückert and Dominique Schröder pp. 17-34	Pairings 2009
Aggregate and Verifiably Encrypted Signatures from Multilinear Maps without Random Oracles Markus Rückert and Dominique Schröder pp. 750-759	ISA 2009

Sanitizable Signatures: How to Partially Delegate Control for
Authenticated Data
Christina Brzuska, Marc Fischlin, Anja Lehmann, and Dominique Schröder
pp. 117-128

BioSig
2009

Mobi: Eine Infrastruktur fuer das Internet der Dinge
Michael Döring and Dominique Schröder

Informatiktag
2006

Patents

Methods for threshold password-hardened encryption and decryption
Russell W.F. Lai, Dominique Schröder, Christoph Egger, Julian Brost · US · 11323248

2022

Doctoral supervision

Davide Li Calsi
Distributed Systems · Ongoing

2025 - present

Ruben Baecker
Privacy-Preserving Access Control · Ongoing

2023 - present

Julian Thomas
Formalizations of privacy-enhancing cryptographic primitives · Ongoing

2023 - present

Paul Gerhart
Fair exchange, optimized Blockchain protocols and privacy · Ongoing

2022 - present

Hien Chu
Privacy-preserving techniques, threshold cryptography · Ongoing

2021 - present

Victoria Ronge
Security and Privacy of Cryptocurrency Signatures · Completed

2018 - 2023

Dominic Deuber
Cryptocurrencies Within The Conflicting Tension Between Law and IT Security · Completed

2017 - 2023

Russel W. F. Lai
Succinct Arguments: Constructions and Applications · Completed

2017 - 2022

Christoph Egger
On Abstraction and Modularization in Protocol Analysis · Completed

2016 - 2022

Sri Aravinda Krishnan Thyagarajan
Cryptographic Locks for Scriptless Cryptocurrency Payments · Completed

2016 - 2021

Guilio Malavolta
Cryptographic Clocks and Applications · Completed

2013 - 2019

Oliver Willers
MEMS Sensors As Physical Unclonable Functions · Completed

2015 - 2019

Tim Ruffing
Cryptography for Bitcoin and Friends · Completed

2014 - 2019

Nils Fleischhacker
Minimal Assumptions in Cryptography · Completed

2012 - 2016

Postdoctoral mentoring

Luigi Russo	07/2025 - present
Dominic Deuber	03/2023 - 06/2024
Subsequent position: IT security expert at the German Accreditation Body	
Mojtaba Khalili	09/2022 - 09/2023
Subsequent position: Assistant Professor, Isfahan University of Technology	
Albrecht Petzoldt	09/2019 - 08/2022
Subsequent position: Dagstuhl	
Bernardo Magri	11/2016 - 11/2018
Subsequent position: Associate Professor at the University of Manchester, UK.	

Teaching

Advanced Privacy Enhancing Technologies	2025
TUW · BSC/MSC	
Data Privacy	2025
FAU · BSC	
IT Security & Law	2025, 2024, 2023
FAU · Master	
Privacy Enhancing Technologies	2025
TUW · BSC	
Blockchain und Kryptowährungen	2024, 2022, 2020
FAU · BSC	
Data Privacy	2024, 2023, 2022
FAU · BSC/MSC	
Introduction to Algorithms	2024, 2023
FAU · BSC	
Introduction to Data Privacy	2023, 2021
FAU · BSC/MSC	
Introduction to Modern Cryptography	2023, 2022, 2021, 2020, 2018, 2017, 2016
FAU · BSC/MSC	
Privacy and Legal Aspects and Their Realizations	2022, 2021
FAU · BSC/MSC	
Privacy-Preserving Cryptocurrencies	2022, 2021, 2020
FAU · BSC/MSC	
Secure Multi-Party Computation	2021, 2017
FAU · BSC/MSC	
Foundations of Privacy	2020
FAU · BSC/MSC	
Password-based Cryptography	2020, 2019, 2018
FAU · BSC/MSC	
Research Oriented Cryptography	2020, 2014
UdS · BSC/MSC	

Cryptocurrencies FAU · BSC/MSC	2019, 2018, 2016
Sabbatical FAU · -	2019
Algorithms and Data Structures FAU · BSC	2018
Foundations of Cryptocurrencies FAU · BSC/MSC	2017
Public-Key Encryption UdS · BSC/MSC	2017, 2015, 2014, 2013, 2012
Cryptocurrencies UdS · BSC/MSC	2016
Secure Multi-Party Computation UdS · BSC/MSC	2016, 2013
Introduction to Modern Cryptography UdS · BSC/MSC	2015, 2014
Recent Advances in Secure Multi-Party Computations UdS · BSC/MSC	2015
Conference leadership	
ACM CCS Area Chair Applied Cryptography	2025, 2026
Chair of IEEE International Conference on Blockchain	2023
Co-Chair of IEEE TEMS on Blockchain and Distributed Ledger Technologies.	2021
General Chair of IACR Theory of Cryptography Conference (TCC).	2019
Poster Session Chair of ACM Conference on Computer and Communications Security (CCS)	2016
Editorial boards	
IACR Communications in Cryptology	2024-2026
ACM Transactions on Privacy and Security	2024
Program committees (selected)	
USENIX Security	2026
ACM Conference on Computer and Communications Security	2019, 2023, 2024-2026
Advances in Cryptology (CRYPTO)	2016, 2019, 2020, 2023
Advances in Cryptology (EUROCRYPT)	2015, 2023, 2027
IEEE Computer Security Foundations Symposium	2020-2024
Advances in Cryptology (ASIACRYPT)	2022

Privacy Enhancing Technologies Symposium (PETS)	2021, 2022, 2024-2027
Public-Key Cryptography (PKC)	2023-2025
The Cryptographers' Track at RSA Conference (CT-RSA)	2025
European Symp. on Research in Computer Security (ESORICS)	2021-2026
Conf. on Cryptology and Network Security (CANS)	2016-2019, 2021-2023
IEEE Security and Privacy (Oakland)	2020, 2021
Advances in Cryptology (INDOCRYPT)	2021
CODASPY	2024
IEEE European Symposium on Security and Privacy (EuroSP)	2018
Asia Conference on Computer and Comm. Security (ASIACCS)	2017, 2018
Conference on Security and Cryptography for Networks (SCN)	2018, 2022
Conference on Applied Cryptography and Network Security (ACNS)	2024
International Conference on Provable Security (PROVSEC)	2016
ACM Conference on Pervasive and Ubiquitous Computing (UbiComp)	2016

Grant reviewing

European Research Council (ERC)
 Deutsche Forschungsgemeinschaft (DFG)
 Deutscher Akademischer Austauschdienst (DAAD)
 Studienstiftung des Deutschen Volkes
 Schloss Dagstuhl
 Österreichische Forschungsförderungsgesellschaft (FFG)
 Natural Sciences and Engineering Research Council of Canada (NSERC)
 Dutch Research Council (NWO)

Journal reviewing

Journal of Cryptology
 Journal of Computer Security
 IET Journal for Information Security
 Journal of Mathematical Cryptology
 ACM Transactions on Privacy and Security (formerly TISSEC)
 IEEE Transactions on Security and Dependable Computing

Doctoral committees

Jenny Ottmann 3rd reviewer · Friedrich-Alexander Universität Erlangen-Nürnberg	2024
Swaroop Ready 2nd reviewer · Indian Institute of Technology, Hyderabad	2024
Victoria Ronge advisor · Friedrich-Alexander Universität Erlangen-Nürnberg	2023
Dominic Deuber advisor · Friedrich-Alexander Universität Erlangen-Nürnberg	2023
Franz-Josef Streit chair of the commission · Friedrich-Alexander Universität Erlangen-Nürnberg	2022
Christoph Egger advisor · Friedrich-Alexander Universität Erlangen-Nürnberg	2022
Russell W. F. Lai advisor · Friedrich-Alexander Universität Erlangen-Nürnberg	2022
Sri Aravinda Krishnan Thyagarajan advisor · Friedrich-Alexander Universität Erlangen-Nürnberg	2021
Markus Zoppelt 1st reviewer · Friedrich-Alexander Universität Erlangen-Nürnberg	2021
Katja Auernhammer 3rd reviewer · Friedrich-Alexander Universität Erlangen-Nürnberg	2021
Kristof Teichel 2nd reviewer · Friedrich-Alexander Universität Erlangen-Nürnberg	2021
Daniel Rausch 2nd Reviewer · Universität Stuttgart	2020
Guilio Malavolta advisor · Friedrich-Alexander Universität Erlangen-Nürnberg	2019
Oliver Willers 1st reviewer, co-advisor · Saarland University	2019
Tim Ruffing co-advisor · Saarland University	2019
Andrea Zimmermann external examiner · Friedrich-Alexander Universität Erlangen-Nürnberg	2018
Nils Fleischhacker advisor · Saarland University	2017
Yan Zhuang chair of the commission · Friedrich-Alexander Universität Erlangen-Nürnberg	2017
Mykolai Protsenko chair of the commission · Friedrich-Alexander Universität Erlangen-Nürnberg	2017
Thomas P. Jakobsen 2nd reviewer · Aarhus University, Denmark	2015
Esfandiar Mohammadi 2nd reviewer · Saarland University	2014

Administrative service

Head of 6 and member of 11 hiring committees	Since 2017
FAU Study commission B. Sc. computer/IT-security	since 2017
Department of computer science board of management FAU	2016-2024
FAU Faculty council of the faculty of engineering (re-elected in 2021)	2019-2024
FAU Department Informatik speaker	2023-2024
FAU Department Informatik Deputy speaker	2021-2023
Study commission computer science	2018-2021
Scientific steering committee of the NCT	2016-2017
Master admission committee (Saarland University)	2012-2016
development of the Bachelor "Cybersicherheit" (Saarland University)	2014

Expert contributions and public service

German Bundestag · Health Committee	2023
Independent expert contribution: Privacy and the use of health data	
German Bundestag · Health Committee	2020
Independent expert contribution: Patient Data Protection Act (PDSG)	
German Bundestag · Health Committee	2019
Independent expert contribution: Digital Healthcare Act (DVG)	

Selected talks and appearances

Privacy of Medical Data	2023
Humboldt Foundation, Network Meeting, Mainz	
Welchen Beitrag leistet die Digitalisierung für eine gerechte Gesundheitsversorgung in Deutschland?	2021
Leopoldina, Berlin, DE.	
Data Privacy for Medical Records	2019
TU Graz, Austria.	
Privacy and Access Control for Outsourced Personal Records.	2016
Leopoldina, Saarbrücken, DE.	
Efficient Pseudorandom Functions via On-the-Fly Adaptation.	2015
Chinese University of Hong Kong (CUHK), Hong Kong.	
Bridging the Gap Between Digital Cryptography and the Physical World.	2014
ATM Security, London, UK.	
UbiCrypt - Die Unterschrift der Zukunft.	2014
Future Talk, CeBiT, Hannover, Germany.	
Security and Privacy in the Internet.	2013
SR Television. Interview with Joachim Weyand	

Verifiable Data Streaming Intel Collaborative Research Institute for Secure Computing, Darmstadt, Germany.	2013
Verifiable Data Streaming University of Maryland, USA.	2013
Verifiable Data Streaming Brown University, USA.	2013
Verifiable Data Streaming IMDEA, Madrid Institute of Advanced Studies, Madrid, Spain.	2013
Provable Security and Beyond. Florida State University, Tallahassee, USA.	2012
History-Free Aggregate Message Authentication Codes. Universite de Rennes, Rennes, France.	2010